CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 9: Matrices

Topic-wise short notes with solved examples

A matrix is a rectangular arrangement of numbers. This chapter covers the order and types of matrices, adding and subtracting them, multiplying by a number, and multiplying two matrices.

1 What a matrix is

- A **matrix** is a rectangular arrangement of numbers in rows and columns, enclosed in brackets. Each number is an **element**.
- A matrix with m rows and n columns has **order** $m \times n$ (read "m by n") and $m \times n$ elements. Rows are counted first.

$$A = \begin{bmatrix} 2 & 5 & -1 \\ 0 & 3 & 4 \end{bmatrix} \text{ has 2 rows and 3 columns, so its order is } 2 \times 3.$$

- Two matrices are **equal** if they have the same order and every pair of corresponding elements is equal.

Type	Description	Example
Row matrix	Only one row	$\begin{bmatrix} 3 & 1 & 4 \end{bmatrix}$
Column matrix	Only one column	$\begin{bmatrix} 2 \\ 7 \end{bmatrix}$
Square matrix	Rows = columns	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
Zero (null) matrix	Every element is 0	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
Identity (unit) matrix I	Square; 1 on the leading diagonal, 0 elsewhere	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

- The **transpose** A^t is obtained by interchanging the rows and columns of A.

Example 1.1 Find x and y if $\begin{bmatrix} x+y & 5 \\ 2 & x-y \end{bmatrix} = \begin{bmatrix} 7 & 5 \\ 2 & 3 \end{bmatrix}$.

SOLUTION

Equal matrices have equal corresponding elements: $x + y = 7$ and $x - y = 3$

Adding: $2x = 10$

$x = 5, y = 2$

2 Addition, subtraction and multiplication by a number

- Two matrices can be added or subtracted only if they have the **same order**. Add or subtract the corresponding elements.
- To multiply a matrix by a number (a **scalar**), multiply every element by that number.
- $A + B = B + A$, and $A + O = A$ where O is the zero matrix of the same order.

Example 2.1 If $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ -2 & 0 \end{bmatrix}$, find $A + B$ and $2A - 3B$.

SOLUTION

$$A + B = \begin{bmatrix} 2+1 & -1+5 \\ 3-2 & 4+0 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & -2 \\ 6 & 8 \end{bmatrix} \text{ and } 3B = \begin{bmatrix} 3 & 15 \\ -6 & 0 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 3 & 4 \\ 1 & 4 \end{bmatrix} \text{ and } 2A - 3B = \begin{bmatrix} 1 & -17 \\ 12 & 8 \end{bmatrix}$$

Example 2.2 Find the matrix X if $X + \begin{bmatrix} 4 & 1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 1 & 5 \end{bmatrix}$.

SOLUTION

$$X = \begin{bmatrix} 6 & 0 \\ 1 & 5 \end{bmatrix} - \begin{bmatrix} 4 & 1 \\ -2 & 3 \end{bmatrix}$$

$$X = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$$

3 Multiplication of matrices

- The product AB exists only if the number of **columns of** A equals the number of **rows of** B.
- If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.
- The element in row i and column j of AB is obtained by multiplying the elements of row i of A by the corresponding elements of column j of B and adding.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$$

Example 3.1 If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$, find AB and BA. Is $AB = BA$?

SOLUTION

$$AB = \begin{bmatrix} 1\times2 + 2\times1 & 1\times0 + 2\times5 \\ 3\times2 + 4\times1 & 3\times0 + 4\times5 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2\times1 + 0\times3 & 2\times2 + 0\times4 \\ 1\times1 + 5\times3 & 1\times2 + 5\times4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 16 & 22 \end{bmatrix}$$

$AB \neq BA$. **Matrix multiplication is not commutative in general.**

Example 3.2 If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$.

SOLUTION

$$A^2 = \begin{bmatrix} 9-1 & 3+2 \\ -3-2 & -1+4 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ -5 & 3 \end{bmatrix}$$

$$5A = \begin{bmatrix} 15 & 5 \\ -5 & 10 \end{bmatrix} \text{ and } 7I = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$A^2 - 5A + 7I = \begin{bmatrix} 8-15+7 & 5-5+0 \\ -5+5+0 & 3-10+7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

4 Properties of multiplication

- In general $AB \neq BA$.
- $A(BC) = (AB)C$ and $A(B + C) = AB + AC$, whenever the products exist.
- $AI = IA = A$ for a square matrix A and the identity matrix I of the same order.
- $AB = O$ does **not** mean that $A = O$ or $B = O$.
- $AB = AC$ does **not** mean that $B = C$.
- $(A + B)^2 = A^2 + AB + BA + B^2$, which is not $A^2 + 2AB + B^2$ unless $AB = BA$.

5 Solving matrix equations

- First find the **order** of the unknown matrix from the orders of the other matrices.
- Write the unknown matrix with letters for its elements, multiply out, and compare corresponding elements.

Example 5.1 Find the matrix X such that $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} X = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$.

SOLUTION

The first matrix is 2×2 and the product is 2×1, so X is 2×1. Let $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

$\begin{bmatrix} 2x + y \\ x + 3y \end{bmatrix} = \begin{bmatrix} 7 \\ 11 \end{bmatrix}$, so $2x + y = 7$ and $x + 3y = 11$

Solving: $x = 2, y = 3$

$$X = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

6 Common mistakes

- Adding matrices of different orders.
- Multiplying corresponding elements to find AB. Use row by column.
- Assuming $AB = BA$, or expanding $(A + B)^2$ as in ordinary algebra.
- Writing the order as columns $\times$ rows.
- Using the number 1 in place of the identity matrix I, or 0 in place of the zero matrix O.

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